NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.2c High-wind forecasting and control for wind turbines

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:35 — 2026-11-08 05:40

Stage conclusion

Project report

A3 High-wind forecasting and control for wind turbines

Objective

Survey existing work and gaps in forecasting and active control of wind turbines under high wind speeds (pitch/yaw/derating, lidar feedforward, tower/blade vibration control), and determine whether there is a viable entry point (upper bounds of feedforward and vibration reduction under extreme/non-stationary inflow).

Current status

On hold (shelved 10-15 07:5x). - One-sentence conclusion: In simulations combining an analytical downburst + synthetic turbulence + offline triggering, using nacelle lidar to trigger shutdown ~7 s early reduces tower-top yaw moment by ~25% and blade-root negative peak moment by ~15% on an off-center path where rapid wind-direction rotation outpaces yaw tracking; however, on a near-center path it worsens the tower-base reverse bending moment by ~14%, failing the pre-specified rule. The pitch feedforward component was already shelved earlier because the toolchain failed the steady-turbulence literature gate. - **Why shelved**: (1) Negative-load worsening on the near-center path exceeds the pre-specified 10% limit; (2) the "pass" on the off-center path carries an unexplained actual yaw overspeed (0.44–0.46°/s vs. 0.3°/s setting) and a LOS turbulence approximation; (3) the physical mechanism is simply "shutting down earlier", whose value depends on identifying downbursts in advance, which is a detection/forecasting problem. - **What is missing**: Realistic downburst inflow (CM1/LES), online triggering, verification of yaw dynamics, gentler shutdown feathering strategies, and a working OpenFAST 5.x lidar feedforward implementation. - **Conditions for restart**: (a) Off-center path gains remain after fixing yaw overspeed, and a shutdown strategy is found that does not worsen tower-base reverse bending moment; (b) the CM1 line or the digital twin wind tunnel line can provide realistic downburst wind fields; (c) a lidar feedforward implementation validated on OpenFAST 5.x becomes available (to restart the pitch feedforward component); (d) there is a concrete wind-farm demand for downburst shutdown. - Migration: Starting 10-15, this node is managed under the "AI weather foundation models" project; experimental records remain in place; old run roots on the workstation have been archived to the supercomputing center.

Boundaries (10-14 18:3x parent project correction)

- Working directory cleanup: the root lead cleaned up the workstation as authorized by the PI; OpenFAST outputs, pilots, logs, and `run_nrel5mw_*` across Segments A/A2/B totaling 8,188 files and 11.39 GB have been moved to the supercomputing center (deleted locally after verifying file counts and byte sizes one by one); `env`, `src`, third-party source code, and build artifacts have been deleted (reconstructible). Future audits will pull from that archive without copying back to the workstation.

Decisions made

- 10-14 kickoff arrangement
- 10-15 verdict of the review by six independent reviewer models: proposal returned
- 10-14 review verdict
- 10-14 DR quota exhausted; submit for review first based on the Grok search plus the lead's check, while running smoke tests in parallel

Attempts and results

Date	What was done	One-line result
10-15	Segment D: corrected turbulence, nacelle LOS 1D, 12 seeds, simple non-lidar triggers S1–S4 (Colab, 360+36 runs)	Near-center: yaw moment reduced by 20.9% but tower-base negative bending moment worsened by 14.4% → failed; off-center: yaw moment reduced by 24–25%, blade-root negative peak improved by 14–15% → calculation passed, but actual yaw overspeed remains unexplained; wind-direction-rate trigger achieved equal load reduction but caused 36/36 false shutdowns → shelved per pre-specified rules
10-15	Review by six independent reviewer models (5 conditional support, 1 not supportive)	Median reduction is real but 1/6 and 2/6 seeds increased instead; tower-base fore-aft negative peak worsened (near-center median 24%, off-center 6/6 seeds), and blade-root negative peak doubled in 2 seeds; turbulence is isotropic and not scaled by translation speed; upstream signal is not a realistic lidar; simple non-lidar triggers not ruled out → proposal returned
10-15	Segment C: lowered thresholds 22/20/18 m/s counterfactual + normal-turbulence false shutdowns (Colab ~0.1 CU)	Lowering the threshold either gave insufficient load reduction or caused 17–18/18 false shutdowns; preview 1D had 10/18 false shutdowns (25 m/s baseline had 9/18) → criterion "preview has unique value"; caveats: the 3 s > 25 m/s rule itself has many false shutdowns, non-monotonic results, and offline triggering
10-15	Segment B shutdown group G0–G4 (96/96, 70 supplemented on Colab, ~3–4 CU)	Pre-specified criteria passed: realistic 1D preview shutdown vs. immediate shutdown reduced median tower-top yaw moment by 15.5%/24.2%, with tower base and blade root essentially unchanged; actual lead time 5.5–9 s, gain mechanism = shutting down earlier; old 26 raw wind boxes no longer exist (process-level provenance confirmed; Segment C also recomputed them independently on Colab)

Date	What was done	One-line result
10-15	Feedforward gate diagnosis: custom-chain time-shift scan + authors' reference chain (Task52, OpenFAST 5.0 channel bridge)	Neither chain passed: best time-shift result was −2.3% (single seed); reference chain rotor-speed std increased by 36–63% instead; BTS ~7 s time offset confirmed by measurement → feedforward component shelved
10-15	Feedforward changed to pitch-rate input into PI integrator + IEC NTM 16 m/s literature gate (3 seeds, Colab ~0.3 CU)	Gate not passed: perfect feedforward rotor-speed std increased by 14–18% instead (literature expects a decrease); dispatched time-shift scan + comparison with authors' reference chain
10-14	Segment B continuation on Colab (single seed 112, G5–G8; recomputed workstation case with 0.000% difference)	G6 realistic 2D −13%, G7 perfect with same lead time +8%, G8 +33% — red flag; the lead diagnosed that the feedforward absolute pitch offset double-counted steady-state pitch, invalidating the 12–24% in A2_guo_v3 as well; ~0.03 CU
10-14	Segment B (partial, 26/96, workstation)	Full-field turbulence + independent upstream preview chain ran through; near-center 3–4 seeds: realistic preview shutdown reduced tower-top yaw moment by ~18–38% but worsened blade-root negative peak, undecided; paused at 23:00 due to C drive space
10-14	Segment A2 supplement: Guo 2023 feedforward law (near-center)	Realistic 1D/2D feedforward reduced tower-base resultant moment by 12%/24%, whereas perfect (2–3 s lead time) achieved only 1.6% — suspected lead-time effect (early derating); pitch feedforward group added back to Segment B to separate lead time from information
10-14	Segment A2: yaw enabled + immediate-threshold shutdown baseline (20/20 runs, 29 min)	Immediate shutdown alone reduced tower-base resultant moment by 15%/37% (non-preview); pitch feedforward ≤1%, dropped; on the off-center path (0.3°/s yaw cannot keep up with 111° wind-direction rotation — physical limitation, judged valid by the lead), preview early shutdown vs. immediate shutdown reduced tower-top yaw moment by 16–67% and blade-root negative peak by 16–41%, with tower-base resultant ≤3%

Date	What was done	One-line result
10-14	Segment A addendum: realistic/perfect gain ratio (v10 exploration)	Pitch feedforward perfect gains mostly <3%, ratios mostly negative or not applicable; shutdown realistic 1D / perfect ≈ 0.8–1.1, but the baseline was the 50 s rule; near-center early shutdown greatly increased tower-base fore-aft negative peak magnitude (−1275%), as shutdown itself induces reverse loading, so A2 must examine both positive and negative peaks
10-14	Segment A: deterministic downburst 24/28 runs (v10)	Pitch feedforward perfect achieved only −0.9%/−3.1%, while realistic increased loads; Case III yaw failure invalidated; shutdown gain suspected to come from eliminating the 50 s wait rather than preview
10-14	Isolated copy of ROSCO 2.10.6 with feedforward injection port added (Apache-2.0)	30 s artificial pulse trial run comparing feedback/feedforward on the same binary gave a maximum pitch difference of 3.9°, confirming the interface works; not a downburst, no load conclusions
10-14	Linux port of research feedforward chain (WRAPPER→LDP→FFP→modified ROSCO 2.6.0) + Nguyen 2012 analytical downburst field	Controller chain 10 s trial run normal; 1D/2D preview in the analytical field advanced the 25 m/s threshold by ~8.6/17.2 s (geometric quantity, not control gain); full-matrix extrapolation exceeded the 24 h gate, no load results produced
10-14	Local smoke test of OpenFAST v5.0.0 + ROSCO 2.10.6 (conda-forge, 60 s simulation completed in 30–38 s wall-clock time)	Toolchain usable; stock ROSCO has no external preview feedforward interface, requiring the MSCA-LIKE (MIT) chain or a custom feedforward term; IEA Task 52 repository has no license and is not used
10-14	Grok quick check + lead verification	Steady-state LAC is mature; no precedent found for preview feedforward under downbursts / gust fronts

Parameter sources

Parameter	Value	Source or "assumption"	Places checked
Shutdown trigger rule	Exceeding 25 m/s for 10 rotor revolutions or 50 s	Nguyen 2012 PDF pp. 112–113	Original text

Parameter	Value	Source or "assumption"	Places checked
Downburst shape function	α=2, c1=0.22, c2=2.75	Nguyen 2012 PDF p. 66 Eqs. 3.2–3.7 (verified by ideas)	Original text
Case I / III parameters	Urm=47, Utrans=12/14, zm0=90/70, rm0=1000/610, y0=−100/−610, etc.	Nguyen 2012 PDF p. 114 Table 4.4 (verified by ideas)	Original text
Ambient wind	12 m/s at 90 m, shear exponent 0.2	Nguyen 2012 PDF p. 83, p. 79 Eq. 3.24	Original text
Turbulence intensity ε	0.1	Nguyen 2012 PDF p. 71 Eq. 3.12	Original text
Yaw rate limit	0.3°/s	Lu 2019 and Nguyen 2012 (cited by reviewer, awaiting ideas check against Jonkman 2009)	To be verified
Pitch rate limit	8°/s	NREL 5 MW (cited by reviewer, awaiting check against Jonkman 2009)	To be verified

Open items and next steps

- Shelved on 10-15; no new trials will be dispatched, and the promotion proposal draft is voided. The audit is now closed: Colab `v4c` recomputation of the legacy 26 cases matched 26/26 across all seven metrics, and on 10-16 the legacy `wind.bts` files in the supercomputing-center archive matched the `v4c` wind boxes byte-for-byte (4/4). No open action items remain.
- Working directory cleanup: the root lead cleaned up the workstation as authorized by the PI; OpenFAST outputs, pilots, logs, and `run_nrel5mw_*` across Segments A/A2/B totaling 8,188 files and 11.39 GB have been moved to the supercomputing center (deleted locally after verifying file counts and byte sizes one by one); `env`, `src`, third-party source code, and build artifacts have been deleted (reconstructible). Future audits will pull from that archive without copying back to the workstation.
- Re-run Deep Research once quota recovers, focusing on confirming "no precedent found".
- The public citation list has been updated with OpenFAST, ROSCO, NREL 5 MW, MSCA-LIKE, and Nguyen 2012/2014.

Existing work and gaps (v2, finalized after review)

Direction	Representative work and page numbers	Maturity	Can we enter?	What is needed
Steady-turbulence lidar feedforward (collective pitch)	Bossanyi 2014 JPCS 555:012011; Guo 2023 WES 8:149 §5.2 (tower-base fore-aft bending moment DEL up to −16.7%, 22 m/s unstable stratification); Schlipf 2018 IEA Task 32	Mature → early commercial	No	—
Feedforward / detection derating under IEC extreme gusts	Schlipf & Raach 2016 JPCS 753:052011 (abstract: significantly reduces tower-base extreme loads); Haizmann 2018 DOI 10.1088/1742-6596/1037/3/032025, Hagemann 2017 DOI 10.18419/opus-9333 (unverified)	Developing	Weak	—
Wind-farm-level preview observation + advance yaw/IPC under gust-front / gust spatiotemporal evolution	Zhang et al. 2026 Energy DOI 10.1016/j.energy.2025.139730 (abstract level only, provided by Reviewer A)	Developing	Pending full-text read	Full text
Wind turbine loads, feedback control, and exceedance shutdown under downbursts	Nguyen 2012 dissertation (PDF pp. 112–113 shutdown rule 50 s; p. 133 / journal p. 116 pitch + yaw insufficient, shutdown too late); Nguyen & Manuel 2014 DOI 10.3390/en7106527; Lu 2019 Energies 12:2773 (LES, five stability cases); Pratapa 2023 DOI 10.1115/1.4055076 (PDF pp. 7–8 including turbulence)	Developing (primarily load assessment)	Yes	Analytical downburst 4D field + OpenFAST

Direction	Representative work and page numbers	Maturity	Can we enter?	What is needed
Downburst dynamics / FSI of already parked (feathered) turbines after warning	Ahmed et al. 2022 DOI 10.1016/j.engstruct.2021.113440; 2023 JSE DOI 10.1061/JSENDH.STENG-11923 (local library 522APTGW p. 3 "warning assumed"); 2023 ES DOI 10.1016/j.engstruct.2023.116779	Developing	No (already done)	—
Single-turbine nacelle lidar preview feedforward under downburst inflow; preview-determined shutdown time T	Not found (searched by Grok, by the lead twice, and by 3 reviewers)	No precedent found (pending DR re-run)	**Yes — entry point of this node**	IEAWindTask52/LidarAssistedControl (modified ROSCO 2.6.0) + 4D field upstream sampling
Forecast-driven wind-farm-level shutdown / derating (hours–minutes)	Jia 2023 Electronics DOI 10.3390/electronics12214443, Li 2022 Atmosphere 13:451 (unverified)	Basic shutdown mature, optimization developing	Medium (not our group's strength)	A1 data fetching
Tower / blade vibration control	Fakhry 2026 review DOI 10.1177/14613484261437475 (unverified)	TMD/IPC mature, active control developing	Medium (shared with A2)	Add TMD to simulation chain

One-sentence conclusion: Worth continuing (narrow scope). **Minimal experiment**: (four-group comparison + T scan, ≥6 seeds × 2 paths, criteria pre-fixed).

Literature basis

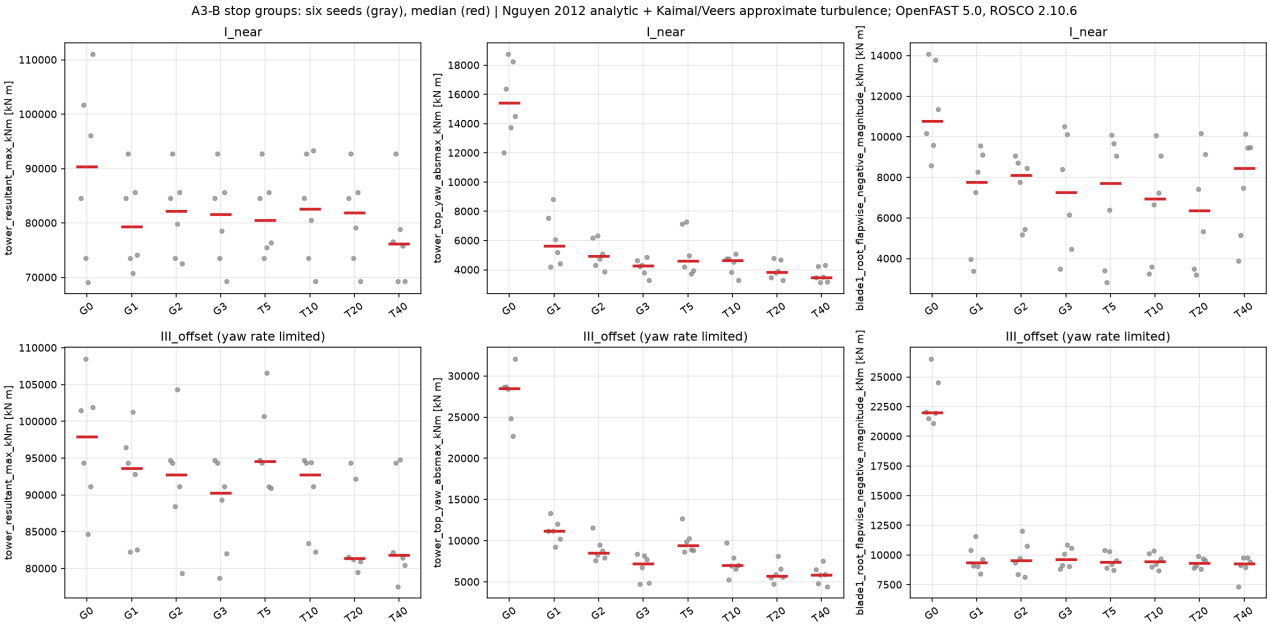
- **A3 thematic index (built by litwiki on 10-14)**: . Cards: Guo 2023 (16.7% is DEL, journal p. 167 / PDF p. 19 §5.2.3; tower / blade-root / low-speed-shaft DEL reductions of 3.0–16.7% / 1.5–6.0% / 1.7–5.0%, journal p. 168); Schlipf & Raach 2016; Nguyen & Manuel 2014; Nguyen 2012 (Table 4.4 thesis p. 97 / PDF p. 114, t0/t1 in min; shutdown rule PDF pp. 112–113); Lu 2019; Haizmann 2018 (perfect 5 s preview, derating to 65% of rated, PDF p. 4); Zhang 2026 metadata card only (results not citable); Pratapa 2023 card Z454567N updated to "mean field + frequency-domain turbulence".
- Existing in knowledge base (downburst × wind turbine, all unread in detail or only skimmed): Pratapa, Nguyen & Manuel 2023 (DOI 10.1115/1.4055076, non-turbulent axisymmetric downburst → 5 MW turbine loads, loads depend on yaw/pitch control state); Nguyen 2012 dissertation "The influence of thunderstorm downbursts on wind turbine design" (unread); Zhang et al. 2015 microburst wind turbine model test; Ji et al. 2025 (attachment identity to be verified).
- Lidar feedforward, wind turbine vibration control: no cards in the library.

Notes from the PI

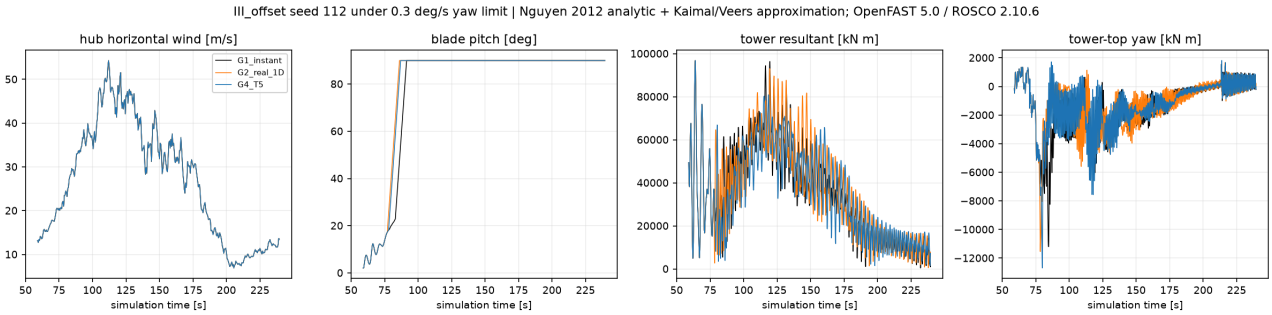
10-22 citation snowballing

- Pool (1,850 papers, 1,556 after initial screening, 90 uniquely from snowballing). Selected 7 papers (optimal control during storms, emergency shutdown loads, comparison of shutdown procedures, yaw/pitch handling under hurricanes, eyewall wind-induced response, etc.) and submitted them to LitWiki Batch 8. Directly relevant to "14% degradation under another operating condition and unresolved yaw anomaly" are the two papers on shutdown procedures and hurricane yaw handling, to be cross-checked once literature cards are built.
- 10-22 Batch 8 verification (#1–7): None overturned this line's findings of "14.4% degradation in reverse bending moment under the near-center operating condition" or "unresolved cause of yaw overspeed." The shutdown-procedure comparison (#3, PDF pp.6–9,15) supports that "shutdown itself can worsen certain loads"; the hurricane yaw/pitch misalignment studies (#4, #5) support verifying yaw and pitch separately, though both prescribe misalignment as an input and cannot explain this line's actuator rate of 0.44–0.46°[local path omitted] They can serve as literature basis for "separately testing yaw and pitch handling" among the R23.2c next-step options.

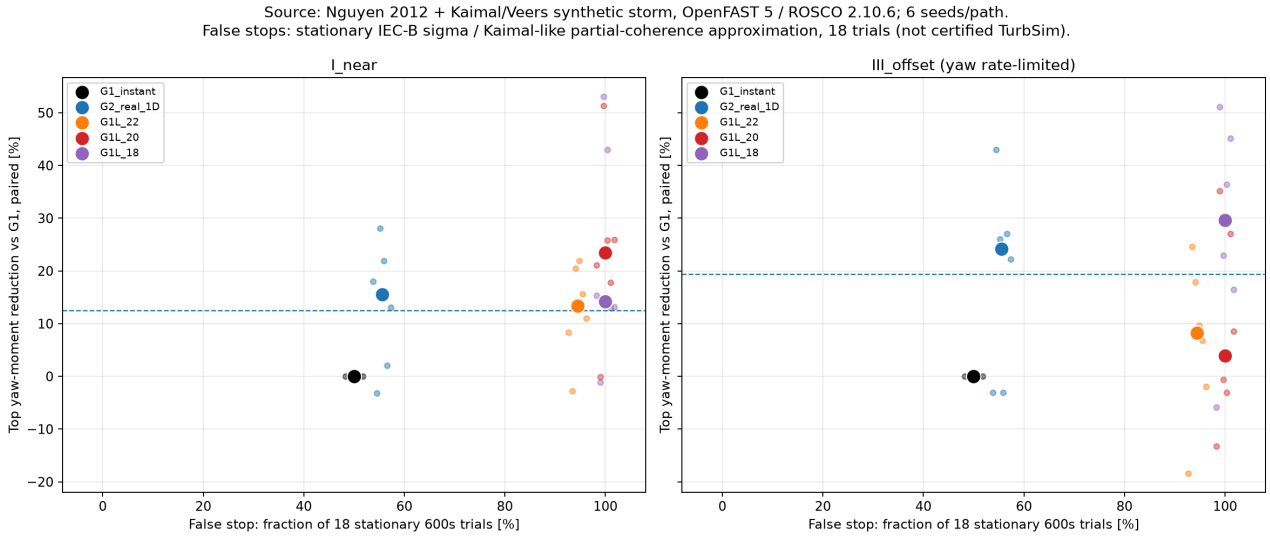
Key figures



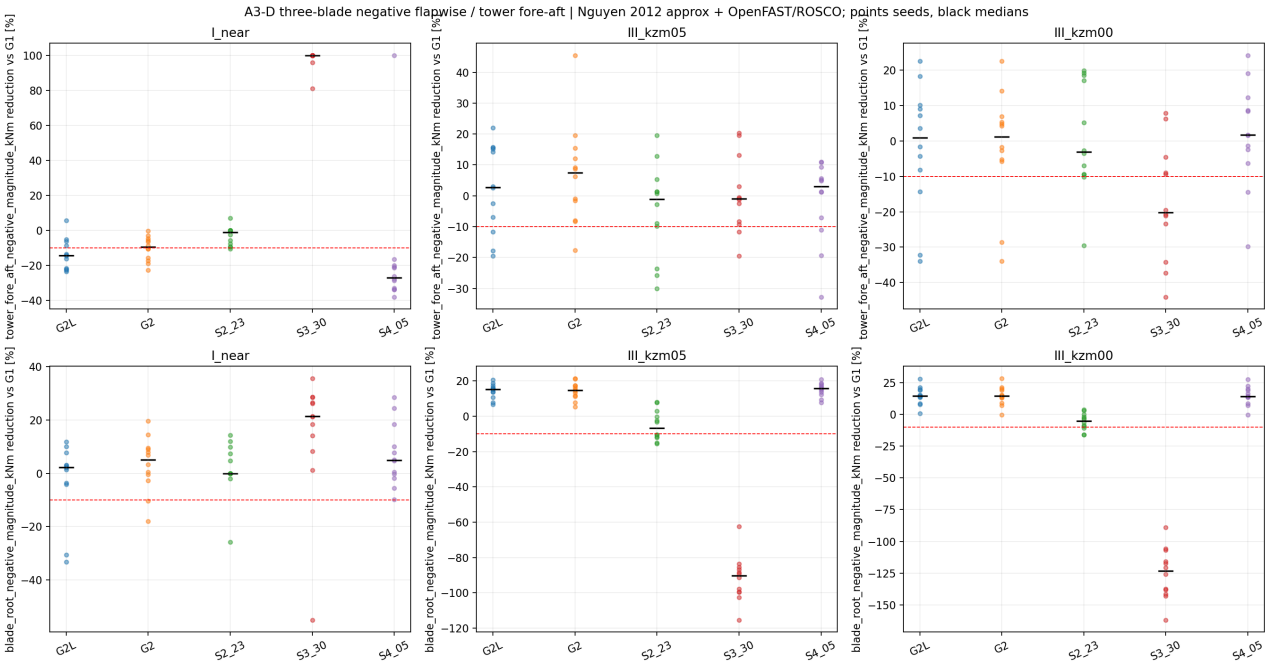
median scatter B stop



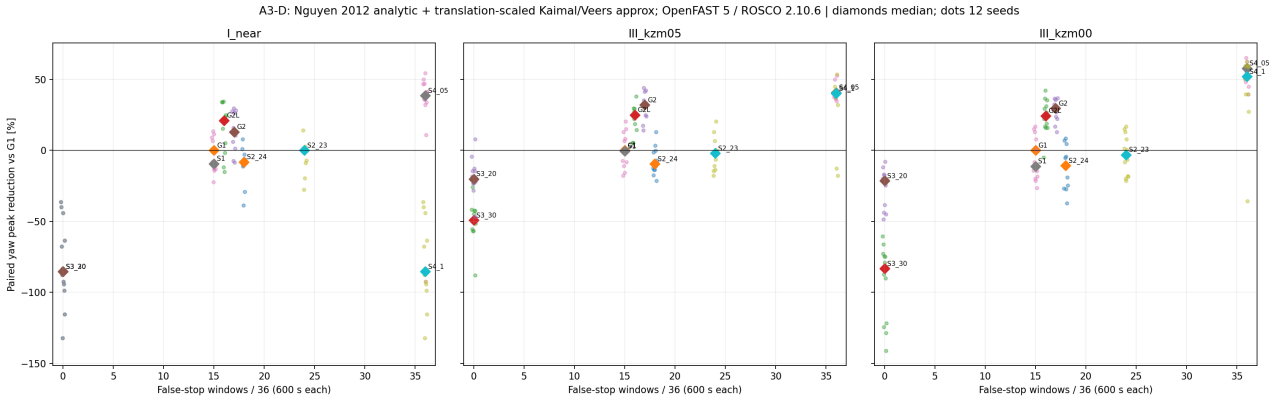
typical seed B stop



yaw vs false stop C



negative load D


A3-D: Nguyen 2012 analytic + translation-scaled Kaimal/Veers approx; OpenFAST 5 / ROSCO 2.10.6 | diamonds median; dots 12 seeds
I_near
III_kzm05
III_kzm00
Paired yaw peak reduction vs G1 [%]
False-stop windows / 36 (600 s each)
False-stop windows / 36 (600 s each)
False-stop windows / 36 (600 s each)
G1
G2
S1
S2_23
S2_24
S3_20
S3_30
S4_05
S4_1

tradeoff D